A Experiment Details

A.1 Effect of Iteration Numbers

The precision, recall, and F1 performance of different LLMs on different datasets with different iteration numbers is shown in Table 6, Table 7, Table 8, and Table 9.

Method	TREC						WebAP					
	listwise			pointwise			listwise			pointwise		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Vanilla	36.82	60.13	45.67	29.92	**91.61**	45.11	13.07	50.83	20.79	13.30	86.29	23.05
UJ-ExpA	48.51	61.15	54.10	28.12	96.27	43.53	18.83	54.16	27.94	14.65	**91.82**	25.27
UJ-ImpA	40.16	60.53	48.29	33.95	83.76	48.31	16.46	52.45	25.06	17.56	73.55	28.35
5-sampling	46.64	59.56	52.31	-	-	-	20.61	56.22	30.16	-	-	-
ITEM-A$_s$ w. ExpA (m=1)	48.07	61.04	53.78	34.21	89.11	49.44	20.57	53.81	29.76	17.86	78.41	29.10
ITEM-A$_s$ w. ExpA (m=2)	50.58	61.86	55.65	35.87	88.73	51.09	21.11	50.85	29.83	18.27	82.00	29.88
ITEM-A$_s$ w. ExpA (m=3)	50.61	59.88	54.86	36.23	90.46	51.74	23.57	48.14	31.65	18.73	81.96	30.50
ITEM-A$_s$ w. ExpA (m=4)	50.01	61.15	55.02	**36.41**	90.36	**51.90**	21.44	44.62	28.96	**19.19**	80.59	**31.00**
ITEM-A$_s$ w. ExpA (m=5)	50.61	59.88	54.86	36.14	90.46	51.65	24.07	47.09	31.86	19.17	78.94	30.85
ITEM-A$_s$ w. ImpA (m=1)	39.97	64.62	49.39	30.98	89.38	46.01	16.88	57.13	26.06	17.10	81.65	28.28
ITEM-A$_s$ w. ImpA (m=2)	43.14	61.52	50.72	30.90	87.00	45.60	19.41	54.82	28.67	18.88	78.06	30.40
ITEM-A$_s$ w. ImpA (m=3)	44.43	62.82	52.05	31.68	87.99	46.59	19.21	54.20	28.36	18.69	77.77	30.13
ITEM-A$_s$ w. ImpA (m=4)	44.72	61.29	51.71	31.66	87.40	46.49	17.44	47.11	25.46	18.95	78.06	30.50
ITEM-A$_s$ w. ImpA (m=5)	44.63	60.98	51.54	31.80	89.32	46.91	18.98	48.88	27.35	19.05	76.69	30.52
ITEM-AR$_s$ (m=1)	43.65	65.34	52.34	-	-	-	25.04	**60.99**	35.50	-	-	-
ITEM-AR$_s$ (m=2)	45.10	65.46	53.40	-	-	-	24.42	51.97	33.23	-	-	-
ITEM-AR$_s$ (m=3)	49.07	**65.96**	56.27	-	-	-	**27.70**	55.95	**37.06**	-	-	-
ITEM-AR$_s$ (m=4)	50.96	62.32	56.07	-	-	-	23.77	53.40	32.90	-	-	-
ITEM-AR$_s$ (m=5)	**53.01**	63.60	**57.82**	-	-	-	25.85	47.56	33.50	-	-	-

Table 6: The utility judgments performance (%) of Mistral on retrieval datasets (Numbers in parentheses represent m-values). Numbers in bold indicate the best performance.

A.2 Quality of Utility Judgments

The relevance labels of TREC DL are of a four-point scale, and we consider the highest level as having utility. To see the utility judgment performance when we consider lower grades to have utility, we measure the precision of utility judgments of Mistral on TREC DL when passages of different grades are treated as positive in Table 10. We can see that almost 70% of the results of positive utility judgments are highly relevant to the question.

A.3 k values in ITEM-A$_r$

Different ranking performance of k values in ITEM-A$_r$ is shown in Table 11. Considering the performance of utility ranking and utility judgments, we set k=5.

A.4 ITEM-AR$_r$

We evaluate two ranking performances of ITEM-AR$_r$ during the same loop, with the experimental results shown in Table 12. We find that under the ITEM-AR$_r$ framework, relevance ranking and utility ranking are both improved, and utility ranking performance is generally better than relevance ranking. However, as seen in Table 3 and Table 12, performing ranking twice in the same iteration may not yield better ranking results than performing utility ranking once in the iteration.

B Case Study

Figure 9 presents two cases from the TREC DL dataset using Mistral under ITEM-A$_s$. For the first question in Figure 9, the first pseudo-answer, though relatively correct, includes irrelevant information, leading to a misjudgment of "Passage-2" as "utility". Based on the results of the first round of utility judgments, the second round of the pseudo-answer is more accurate and free of irrelevant content. Consequently, all three passages in the second round of utility judgments have utility in answering the question. For the second

Method	TREC						WebAP					
	listwise			pointwise			listwise			pointwise		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Vanilla	34.67	**85.80**	49.39	31.42	**98.47**	47.64	12.69	77.15	21.79	14.65	**87.36**	25.09
UJ-ExpA	39.21	80.98	52.83	38.27	90.15	53.73	16.32	77.92	26.99	18.04	77.15	29.25
UJ-ImpA	33.92	83.36	48.22	38.68	71.47	50.20	15.57	**82.79**	26.22	17.22	47.61	25.29
5-sampling	39.04	80.98	52.68	-	-	-	17.52	83.49	28.97	-	-	-
ITEM-A$_s$ w. ExpA (m=1)	39.68	82.88	53.66	37.58	84.84	52.09	17.54	63.67	27.50	19.65	74.31	31.08
ITEM-A$_s$ w. ExpA (m=2)	**42.35**	84.77	**56.48**	38.25	84.58	52.68	17.39	60.25	26.99	20.23	73.01	31.68
ITEM-A$_s$ w. ExpA (m=3)	42.00	84.15	56.03	37.84	85.50	52.46	19.12	62.87	29.32	**20.91**	74.63	32.67
ITEM-A$_s$ w. ExpA (m=4)	41.85	84.41	55.96	38.12	85.16	52.67	17.53	61.85	27.31	20.44	73.83	32.02
ITEM-A$_s$ w. ExpA (m=5)	42.36	84.15	56.35	37.35	84.69	51.84	18.94	62.87	29.12	20.88	75.45	**32.71**
ITEM-A$_s$ w. ImpA (m=1)	39.63	83.42	53.73	39.70	82.87	53.68	15.48	73.66	25.59	20.04	64.06	30.53
ITEM-A$_s$ w. ImpA (m=2)	38.75	85.63	53.35	38.15	82.36	52.14	15.50	76.47	25.77	18.54	62.69	28.62
ITEM-A$_s$ w. ImpA (m=3)	40.84	84.86	55.14	40.58	79.64	53.76	15.99	70.99	26.10	19.54	61.32	29.64
ITEM-A$_s$ w. ImpA (m=4)	38.88	82.74	52.90	39.34	81.74	53.12	15.03	74.41	25.01	19.72	59.95	29.68
ITEM-A$_s$ w. ImpA (m=5)	41.26	84.61	55.47	**40.92**	82.14	**54.63**	15.49	68.93	25.29	19.84	57.21	29.46
ITEM-AR$_s$ (m=1)	34.53	84.17	48.97	-	-	-	**20.05**	72.88	**31.44**	-	-	-
ITEM-AR$_s$ (m=2)	36.27	83.19	50.51	-	-	-	15.92	79.01	26.50	-	-	-
ITEM-AR$_s$ (m=3)	38.04	82.68	52.10	-	-	-	17.93	76.87	29.08	-	-	-
ITEM-AR$_s$ (m=4)	37.28	83.70	51.58	-	-	-	16.60	78.81	27.42	-	-	-
ITEM-AR$_s$ (m=5)	40.25	81.37	53.86	-	-	-	17.04	74.83	27.75	-	-	-

Table 7: The utility judgments performance (%) of Llama 3 on retrieval datasets (Numbers in parentheses represent m-values). Numbers in bold indicate the best performance.

Method	TREC						WebAP					
	listwise			pointwise			listwise			pointwise		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Vanilla	42.13	**79.98**	55.19	33.86	94.40	49.84	17.13	**83.45**	28.43	15.80	**89.42**	26.85
UJ-ExpA	45.74	77.36	57.49	32.06	**96.19**	48.09	19.51	69.86	30.50	16.23	88.74	27.44
UJ-ImpA	44.19	77.11	56.18	33.45	90.36	48.83	18.37	80.14	29.89	15.58	84.51	26.32
5-sampling	50.78	74.77	60.49	-	-	-	20.70	65.83	31.49	-	-	-
ITEM-A$_s$ w. ExpA (m=1)	55.55	71.48	62.52	37.83	91.94	53.61	26.74	59.45	36.89	19.73	84.95	32.02
ITEM-A$_s$ w. ExpA (m=2)	57.95	70.40	63.57	40.74	93.04	56.67	29.43	60.58	39.62	19.62	78.62	31.40
ITEM-A$_s$ w. ExpA (m=3)	58.36	68.88	63.18	40.00	91.88	55.74	29.30	60.91	39.57	19.80	76.20	31.43
ITEM-A$_s$ w. ExpA (m=4)	**58.48**	70.67	**64.00**	40.25	93.38	56.25	29.11	61.03	39.42	20.48	79.63	32.58
ITEM-A$_s$ w. ExpA (m=5)	58.34	69.69	63.51	39.29	92.16	55.09	29.76	60.68	**39.93**	20.58	80.42	**32.77**
ITEM-A$_s$ w. ImpA (m=1)	54.36	65.08	59.24	40.89	82.20	54.61	24.79	64.37	35.80	18.78	67.00	29.34
ITEM-A$_s$ w. ImpA (m=2)	55.88	63.11	59.27	43.32	83.13	**56.96**	27.68	62.03	38.28	20.70	70.54	32.00
ITEM-A$_s$ w. ImpA (m=3)	57.33	64.17	60.56	41.66	80.48	54.90	**30.01**	63.60	40.78	21.51	66.77	32.54
ITEM-A$_s$ w. ImpA (m=4)	55.98	62.24	58.95	**42.34**	80.65	55.53	28.43	60.11	38.60	20.60	65.63	31.36
ITEM-A$_s$ w. ImpA (m=5)	56.63	62.19	59.28	41.49	83.57	55.45	29.05	60.66	39.29	**21.51**	68.03	32.68
ITEM-AR$_s$ (m=1)	51.94	76.90	62.00	-	-	-	25.32	65.84	36.58	-	-	-
ITEM-AR$_s$ (m=2)	53.77	76.19	63.05	-	-	-	25.55	59.26	35.70	-	-	-
ITEM-AR$_s$ (m=3)	52.41	74.04	61.37	-	-	-	27.61	63.96	38.58	-	-	-
ITEM-AR$_s$ (m=4)	52.75	73.78	61.52	-	-	-	28.84	61.85	39.34	-	-	-
ITEM-AR$_s$ (m=5)	52.77	76.28	62.39	-	-	-	28.76	62.54	39.40	-	-	-

Table 8: The utility judgments performance of ChatGPT on retrieval datasets (Numbers in parentheses represent m-values). Numbers in bold indicate the best performance.

References of RAG	Mistral		Llama 3		ChatGPT	
	EM	F1	EM	F1	EM	F1
Golden Evidence	46.09	62.59	64.45	76.64	66.40	76.86
RocketQAv2	31.58	47.69	**50.96**	62.01	46.54	57.00
Vanilla	31.16	47.43	49.09	60.56	48.52	58.64
UJ-ExpA	32.76	48.46	49.63	61.10	47.72	58.01
UJ-ImpA	30.67	46.83	48.88	60.26	49.01	59.30
5-sampling	33.24	48.84	48.72	60.71	48.90	58.97
ITEM-A$_s$ w. ExpA (m=1)	32.98	49.00	50.16	61.88	49.38	59.78
ITEM-A$_s$ w. ExpA (m=2)	**34.31**	**50.08**	50.48	**62.32**	49.22	59.99
ITEM-A$_s$ w. ExpA (m=3)	33.73	49.63	50.27	62.09	**49.69**	**60.18**
ITEM-A$_s$ w. ExpA (m=4)	34.21	50.07	50.43	62.20	-	-
ITEM-A$_s$ w. ExpA (m=5)	33.78	49.63	50.27	62.07	-	-
ITEM-A$_s$ w. ImpA (m=1)	32.17	48.51	50.37	61.89	48.75	58.99
ITEM-A$_s$ w. ImpA (m=2)	32.49	48.67	49.63	61.16	49.11	59.14
ITEM-A$_s$ w. ImpA (m=3)	32.39	48.47	49.68	61.48	48.69	58.94
ITEM-A$_s$ w. ImpA (m=4)	32.71	48.84	49.41	61.03	-	-
ITEM-A$_s$ w. ImpA (m=5)	32.33	48.44	49.73	61.42	-	-
ITEM-AR$_s$ (m=1)	33.30	49.26	50.27	61.69	49.52	59.64
ITEM-AR$_s$ (m=2)	33.57	49.16	50.70	61.92	49.01	59.75
ITEM-AR$_s$ (m=3)	33.40	49.27	49.36	60.97	49.06	59.67
ITEM-AR$_s$ (m=4)	33.46	49.24	49.84	61.54	-	-
ITEM-AR$_s$ (m=5)	33.89	49.58	49.20	60.84	-	-

Table 9: The answer generation performance (%) of all LLMs in the listwise approach. Numbers in bold indicate the best performance except the answer performance using golden evidence. Due to the high cost of using ChatGPT, we only tested with m=1,2,3 on ChatGPT.

m	label$\geq$1	label$\geq$2	label$\geq$3
m=1	82.08	68.34	48.07
m=2	83.86	69.53	50.58
m=3	84.23	**71.06**	**50.61**
m=4	**84.63**	70.18	50.01
m=5	84.52	70.69	50.61

Table 10: The precision scores (%) of utility judgments using Mistral in different m (iteration) values. "label" is the manual annotation in the original dataset, i.e., [3]: Perfectly relevant; [2]: Highly relevant; [1]: Related; [0]: Irrelevant.

k, m	Ranking					Utility judgments		
	N@1	N@3	N@5	N@10	N@20	P	R	F1
k=1, m=1	72.76	71.27	70.57	72.69	84.08	53.66	24.09	33.25
k=1, m=2	76.02	71.54	71.38	73.66	84.78	58.54	28.73	38.54
k=1, m=3	77.24	72.83	71.83	73.87	85.20	**59.76**	28.84	38.90
k=1, m=4	77.24	73.04	71.91	73.90	85.25	59.76	28.84	38.90
k=1, m=5	76.02	72.11	71.42	73.45	84.98	58.54	28.71	38.53
k=5, m=1	72.76	71.27	70.57	72.69	84.08	33.17	57.31	42.02
k=5, m=2	78.46	73.74	72.86	75.48	86.09	32.93	58.37	42.10
k=5, m=3	79.27	75.00	74.27	75.78	86.80	34.15	62.57	44.18
k=5, m=4	79.67	75.92	**75.35**	**76.83**	**87.23**	35.12	61.40	**44.68**
k=5, m=5	79.67	75.32	74.61	76.20	86.82	34.63	61.25	44.25
k=10, m=1	72.76	71.27	70.57	72.69	84.08	22.56	68.03	33.88
k=10, m=2	78.05	72.64	72.90	75.48	85.74	23.66	75.47	36.02
k=10, m=3	**80.89**	**76.58**	74.54	76.30	86.94	23.78	**75.65**	36.19
k=10, m=4	78.05	74.70	72.85	75.12	85.72	24.51	74.17	36.85
k=10, m=5	79.67	75.60	74.84	76.54	86.88	23.66	74.42	35.90

Table 11: The utility ranking performance and utility judgments performance of Mistral on TREC DL dataset in ITEM-A$_r$. "N@k" means "NDCG@k". Numbers in bold indicate the best performance.

m	NDCG@5	NDCG@10	NDCG@20	Utility-F1
1	71.29 / 72.77	72.90 / 74.96	84.56 / 85.75	43.13
2	**72.54** / 70.99	74.81 / 73.76	85.77 / 85.28	40.21
3	72.07 / **74.14**	74.14 / **76.63**	85.53 / **86.57**	**45.67**
4	71.02 / 71.06	74.30 / 74.03	85.09 / 85.16	43.82
5	72.26 / 70.12	**75.83** / 72.59	**85.88** / 84.77	44.10

Table 12: Ranking of topical relevance and utility judgments performance (%) of ITEM-AR$_r$ using Mistral on the TREC DL dataset. "a/b" means "relevance ranking performance / utility ranking performance". Numbers with underline mean better performance among all variants of ITEM with the same m.

question in Figure 9, the first pseudo-answer is correct, but two passages without utility are judged as "utility". The second pseudo-answer, with slight rewording, results in all selected passages being correct.

C Datasets and Evaluation

Dataset	#Psg	#PsgLen	#Q	#Rels/Q
TREC	8.8M	93	82	212.8
WebAP	379k	45	73	76.4
NQ	21M	100	1868	1.0
GTI-NQ	10	100	1863	1.0

Table 13: Statistics of experimental datasets.

Detailed statistics of the experimental datasets are shown in Table 13. We use the *trec_eval* official tool for evaluation of NDCG.

D Answer Passage Retrieval

Non-factoid questions are usually expected longer answers, such as sentence-level or passages-level (Keikha et al., 2014a; Yang et al., 2016; Keikha et al., 2014b). Yang et al. (2016) developed an annotated dataset for answer passage retrieval called WebAP, which has an average of 76.4 qrels per query. Cohen et al. (2018) and Hashemi et al. (2020) introduced the WikiPassageQA dataset and ANTIQUE dataset for answer passage retrieval, respectively. Compared to the WebAP dataset, WikiPassageQA and ANTIQUE have incomplete annotations, with an average of 1.7 qrels and 32.9 qrels per query (Hashemi et al., 2019,

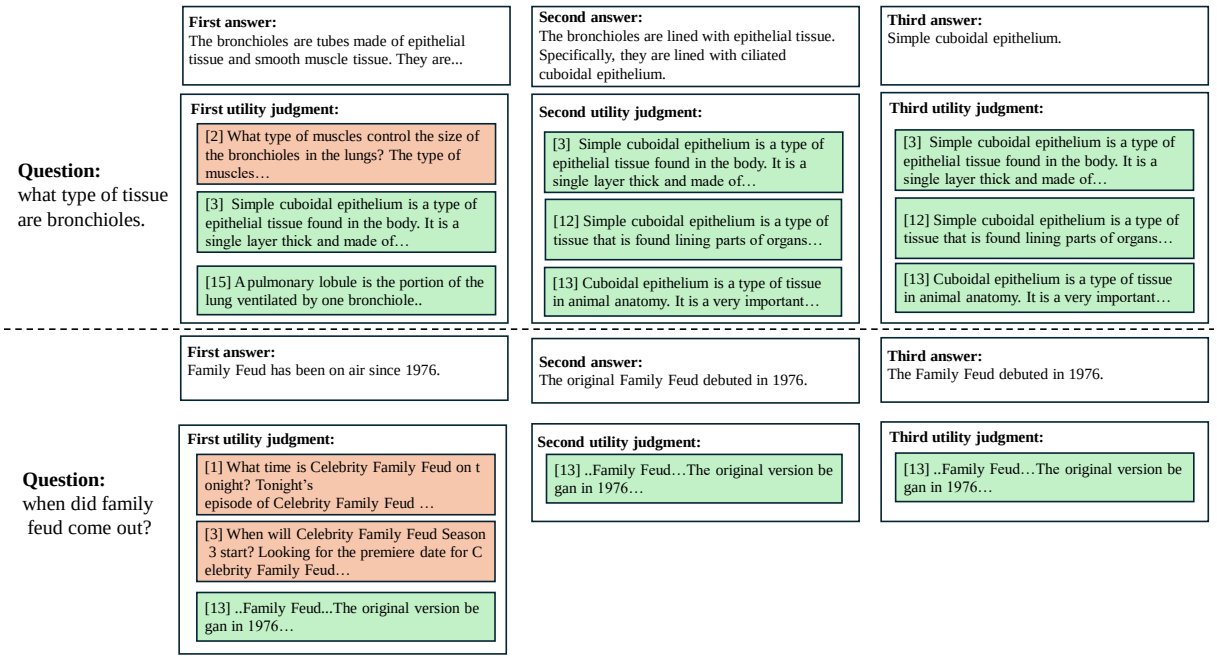

Figure 9: An example of our ITEM-A$_s$ using Mistral on the TREC dataset. Green means the passage has utility, and orange means the passage does not have utility.

2020). Bi et al. (2019) created the PsgRobust dataset for answer passage retrieval, which is built on the TREC Robust collection (Voorhees et al., 2003) following a similar approach to WebAP but without manual annotation.

E k-sampling

The output of k-sampling each time contains explicit answers and utility judgments. If the question length is l_q, the total length of the input passages is l_p, and the average length of a single passage is l_{avg}, then the k-sampling input cost is $(k+1) \times (l_q + l_p)$. If the average length of the output explicit answer is l_a, and the average length of the output utility judgments is l_u, then the k-sampling output cost is $(k+1) \times (l_a + l_u)$. Taking ITEM-As as an example, with a maximum of three iterations, the maximum input cost for utility judgments is $3 \times (l_q + l_p)$. For answer generation, the longest input is $l_q + l_p$ and the shortest is $l_q + l_{\text{avg}}$. Therefore, the maximum input cost for ITEM-As is $6 \times (l_q + l_p)$ and the minimum is $4 \times (l_q + l_p) + 2 \times (l_q + l_{\text{avg}})$. The maximum output cost is $3 \times (l_a + l_u)$. Therefore, in order to ensure fairness in the calculation of large language model parameters, we choose k=5.

F Retrievers

We use two representative retrievers to gather candidate passages in $\mathcal{D}$ for utility judgments. Following with previous works (Zhang et al., 2024; Sun et al., 2023), we employ RocketQAv2 (Ren et al., 2021) and BM25 (Robertson et al., 2009) for the NQ dataset and retrieval datasets(i.e., TREC DL and WebAP datasets), respectively. Based on the retrieval results to build the $\mathcal{D}$ we have two settings: (i) For TREC DL and WebAP datasets, we select the top-20 BM25 retrieval results. If these do not include passages with utility, we replaced the last one with a utility-annotated passage. (ii) For the NQ dataset, we use the top-10 dense retrieval results to form the candidate list $\mathcal{D}$, following the GTU setting of Zhang et al. (2024).

G Inference Efficiency

Table 14 shows more analysis of Mistral on the TREC DL dataset using the listwise input form. The temperature of the LLMs is set to 0 during the generation process, and we used a single run. Due to the iterations, the average input token length of our methods is relatively large. The cost of ITEM-A$_s$ is about 0.5 times that of the 5-sampling, and ITEM-AR$_s$ is about 1.5 times the cost of 5-sampling. Our

Methods	#IT/Q	#OT/Q	#RT(s)/Q
Vanilla	2507	23	0.49
UJ-ExpA	2529	59	0.94
UJ-ImpA	2533	41	0.71
5-sampling	12647	296	4.72
ITEM-A_s(m=1)	4628	47	0.96
ITEM-A_s(m=3)	10603	154	2.13
ITEM-AR_s(m=1)	7107	211	3.15
ITEM-AR_s(m=3)	18224	624	8.61

Table 14: An empirical analysis of the actual cost of baselines and our ITEM. "IT", "OT", and "RT" mean input tokens, output tokens, and run time, respectively.

framework provides a way of automatically obtaining high-quality labeled data for each task in RAG. These annotations can be used to train regular RAG models. The cost should be worth it (much better than k-sampling).

H Instruction Details

H.1 Instruction of Listwise and Pointwise Approaches

For the prompts of the NQ dataset using ChatGPT, we follow the setting of Zhang et al. (2024), otherwise, we use the following prompts. Following Sun et al. (2023), we input N passages using the form of multiple rounds of dialogue in the listwise approach. The prompts we used in our experiments are shown in Figure 10 and Figure 11.

user:
You are the utility judge, an intelligent assistant that can select the passages that have utility in answering the question.
assistant:
Yes, i am the utility judge.
user:
I will provide you with {num} passages, each indicated by number identifier [].
I will also provide you with a reference answer to the question.
Select the passages that have utility in generating the reference answer to the following question from the {num} passages: {query}.
assistant:
Okay, please provide the passages and the reference answer.
user:
[1] {{passage_1}}
assistant:
Received passage [1]
user:
[2] {{passage_2}}
assistant:
Received passage [2]
(more passages) ...
user:
Question: {query}.
Reference answer: {answer}.
The requirements for judging whether a passage has utility in answering the question are: The passage has utility in answering the question, meaning that the passage not only be relevant to the question, but also be useful in generating a correct, reasonable and perfect answer to the question. Directly output the passages you selected that have utility in generating the reference answer to the question. The format of the output is: 'My selection:[[i],[j],...].'. Only response the selection results, do not say any word or explain.

Figure 10: Instruction in the listwise approach.

H.2 Instruction of the Ranking Approach

For RankGPT, we directly use the instruction of Sun et al. (2023) for relevance ranking, as shown in Figure 14. For the relevance ranking in our ITEM, the instructions are shown in Figure 12 and Figure 13.

H.3 Instruction of Answer Generation

Li et al. (2023) utilize LLM to generate the missing information in the provided documents for the current question and then re-retrieve it as relevant feedback. Therefore, we have also designed two kinds of pseudo answers for utility judgments, i.e., (i) the explicit answer, which produces an answer based on the given information, and (ii) the implicit answer, which does not answer the question directly but gives the information necessary to answer the question. The two instructions are shown in Figure 15 and Figure 16.

user:
You are the utility judger, an intelligent assistant that can judge whether a passage has utility in answering the question or not.

assistant:
Yes, i am the utility judger.

user:
I will provide you with a passage and the reference answer to the question. \n Judge whether the passage has utility in generating the reference answer to the following question or not: {query}.

assistant :
Okay, please provide the passage and the reference answer to the question.

user:
Question: {query}.
Reference answer: {answer}.
Passage: {passage}
The requirements for judging whether a passage has utility in answering the question are: The passage has utility in answering the question, meaning that the passage not only be relevant to the question, but also be useful in generating a correct, reasonable and perfect answer to the question.
The reference answer may not be the correct answer, but it provides a pattern of the correct answer. Directly output whether the passage has utility in generating the reference answer to the question or not. If the passage has utility in generating the reference answer, output 'My judgment: Yes, the passage has utility in answering the question.'; otherwise, output 'My judgment: No, the passage has no utility in answering the question.'.

Figure 11: Instruction in the pointwise approach.

user:
You are RankGPT, an intelligent assistant that can rank passages based on their relevance to the query.

assistant:
Yes, i am RankGPT.

user:
I will provide you with {num} passages, each indicated by number identifier []. I will also give you a reference answer to the query. Rank the passages based on their relevance to query: {query}.

assistant :
Okay, please provide the passages and the reference answer.

user:
[1] {{passage_1}}

assistant :
Received passage [1]

user:
[1] {{passage_2}}

assistant :
Received passage [2]

(more passages) ...
user:
Query: {query}.
Reference answer: {answer}
Rank the {num} passages above based on their relevance to the query. The passages should be listed in descending order using identifiers. The most relevant passages should be listed first. The output format should be [] > [] > [] > ..., e.g., [i] > [j] > [k] > ...
Only response the ranking results, do not say any word or explain.

Figure 12: Instruction of the relevance ranking approach in our ITEM.

user:
You are RankGPT, an intelligent assistant that can rank passages based on their utility in generating the given reference answer to the question.
assistant:
Yes, i am RankGPT.
user:
I will provide you with {num} passages, each indicated by number identifier []. I will also give you a reference answer to the question.
Rank the passages based on their utility in generating the reference answer to the question: {query}.
assistant :
Okay, please provide the passages and the reference answer.
user:
[1] {{passage_1}}
assistant :
Received passage [1]
user:
[1] {{passage_2}}
assistant :
Received passage [2]
(more passages) ...
user:
Question: {query}.
Reference answer: {answer}
Rank the {num} passages above based on their utility in generating the reference answer to the question. The passages should be listed in utility descending order using identifiers. The passages that have utility in generating the reference answer to the question should be listed first. The output format should be [] > [] > [] > ..., e.g., [i] > [j] > [k] > ... Only response the ranking results, do not say any word or explain.

Figure 13: Instruction of the utility ranking approach in our ITEM.

user:
You are RankGPT, an intelligent assistant that can rank passages based on their relevance to the query.
assistant:
Yes, i am RankGPT.
user:
I will provide you with {num} passages, each indicated by number identifier [].
Rank the passages based on their relevance to query: {query}.
assistant :
Okay, please provide the passages.
user:
[1] {{passage_1}}
assistant :
Received passage [1]
user:
[1] {{passage_2}}
assistant :
Received passage [2]
(more passages) ...
user:
Query: {query}.
Rank the {num} passages above based on their relevance to the query. The passages should be listed in descending order using identifiers. The most relevant passages should be listed first. The output format should be [] > [] > [] > ..., e.g., [i] > [j] > [k] > ... Only response the ranking results, do not say any word or explain.

Figure 14: Instruction of the ranking approach in Sun et al. (2023).

user:
You are a faithful question and answer assistant. Answer the question based on the given information with one or few words/sentences without the source.
assistant:
Yes, i am the faithful question and answer assistant.

user:
Given the information:
{passage}
Answer the following question based on the given information with one or few words/sentences without the source.
Question: {question}
Answer:

Figure 15: Instruction of the explicit answer generation.

user:
You are a faithful question and answer assistant. Given a question and references. To answer the question, output which information is necessary to answer the question based on the references.
assistant:
Yes, i am the faithful question and answer assistant.

user:
References: {pas}
Question: {question}
To answer the question, output which information is necessary to answer the question based on the references. Do not mention references when printing out necessary information. The format of the output is: 'Necessary information: [xxx]'.

Figure 16: Instruction of the implicit answer generation.